

Dataset comparison for leaf descriptor extraction

Introduction

When measuring leaf traits -such as aspect ratio, margin serration, color and venation- there are some relevant features from a dataset that may make the descriptors extraction easier. The criterion followed in this project compares ten different characteristics among the Flavia, Pl@ntNet300k, PlantCLEF and Herbarium 2021 datasets.

These ten features are the number of images each dataset contains, the type of the images, what species have been included, the capture conditions, leaf availability, background, scale, resolution, annotations and licence/access.

Comparison

	Flavia	Pl@ntNet-300K	PlantCLEF (2015–16 edition)	Herbarium 2021
Images	1,907	306,146 (244k/31k/31k split)	113,205; other editions range from ≈ 1.4 M to ≈ 4 M	2,500,779
Image type	Single isolated leaf, scanned	Field photographs (Pl@ntNet citizen science)	Field photographs plus leaf scans on a uniform background	Digitised herbarium sheets
Species / taxa	32 species (Yangtze delta, China)	1,081 species	$\approx 1,000$ species (France and neighbouring regions)	64,500 taxa, normalised to LCVF
Capture conditions	Controlled and uniform	Uncontrolled: variable light, angle, distance; occlusions	Mixed: uncontrolled for photographs, controlled for the LeafScan subset	Semi-controlled, but pressed and aged material
Leaf availability	100%, one leaf per image, no petiole	Multi-organ; a per-image organ tag allows a leaf subset to be selected	Multi-organ, with explicit <i>Leaf</i> and <i>LeafScan</i> view types: 13,367 + 12,605 training images	Usually present, but overlapping, folded or fragmented
Background	Uniform white	Natural and arbitrary	Natural for photographs; uniform for LeafScan	Uniform pale sheet; labels automatically blurred
Scale	None	None	None	Yes: millimetre ruler and colour chart on the sheet
Resolution	$\approx 1600 \times 1200$ px	570×570 px on average (180–900)	Variable, typical digital-camera sizes	$\approx 6000 \times 4000$ px originals, released at 1000 px
Annotations	Species label only	Species (expertise-weighted user votes), organ, author, licence; official splits	Species, genus, family, view type, observation ID, user quality vote, date, locality, author	Taxon, family, order; accepted / unresolved status
Licence / access	Free download (SourceForge), cite Wu <i>et al.</i> (2007)	CC BY 4.0 (Zenodo)	Creative Commons; download from the Pl@ntNet / LifeCLEF servers after registration	Kaggle (FGVC8), public after registration

PlantCLEF in detail

PlantCLEF is not one dataset but a series of yearly editions whose task changes substantially, so the choice of edition matters more than the name:

- **2014–2016 (Pl@ntView).** Around 1,000 species from France and neighbouring regions, with a view type annotated per image: Branch, Entire, Flower, Fruit, Stem, *Leaf* and *LeafScan*. In the 2015 release this gives 13,367 leaf photographs and 12,605 leaf scans in the training set alone. These are the editions relevant here.
- **2017.** Scaled up with noisy web-crawled images; organ annotation becomes less reliable.
- **2020–2021.** Cross-domain task: identify field photographs from herbarium sheets. Interesting as a precedent for Herbarium 2021, not as a source of leaf images.
- **2022–2023.** Global scale, $\approx 80,000$ species, combining expert-verified and semi-reviewed web data. Diversity, but no gain in measurable leaves.
- **2024–2025.** Multi-label identification in 0.5 m^2 vegetation quadrats, where leaves overlap and cannot be isolated. Not usable for measurement, although the DINOv2 model released with the 2024 edition may be useful as a feature extractor.

A reasonable leaf subset can therefore be selected from PlantCLEF, and the *LeafScan* category is the interesting part: leaves photographed or scanned against a uniform background, i.e. Flavia-like conditions but with $\approx 1,000$ species instead of 32. That gives an intermediate step between the controlled and the natural setting rather than a jump straight from one to the other.

PlantCLEF or Pl@ntNet-300K for the natural-conditions stage?

Both provide a way to select leaves, and neither provides a scale, so the decision rests on what each allows us to *evaluate*:

- **Repeatability.** PlantCLEF groups images by observation, so several images of the same individual plant can be compared. That makes it possible to measure whether a descriptor returns consistent values for the same leaf under different capture conditions — arguably the central question when evaluating descriptor extraction in the field. Pl@ntNet-300K does not expose this grouping.
- **Image quality.** PlantCLEF carries a user quality vote per image, which allows the evaluation set to be stratified by quality instead of being filtered by hand.
- **Controlled tier.** The LeafScan subset has no equivalent in Pl@ntNet-300K.
- **Scale and convenience.** Pl@ntNet-300K is larger (306k images, 1,081 species), has official splits, a single CC BY 4.0 licence and a one-step download, which makes it the more practical option and the better one if a learned segmentation step has to be trained.

The two are complementary rather than exclusive: Pl@ntNet-300K as the volume source for field images, PlantCLEF 2015–16 for the repeatability and image-quality analyses, and its LeafScan subset as an intermediate validation step. Confirming this requires two counts that are not published and that we should compute when downloading: how many Pl@ntNet-300K images carry the `leaf` organ tag and how they are distributed per species, and how many PlantCLEF observations contain more than one leaf image of the same plant.

Proposed selection

- **Flavia** — controlled environment in which to develop and initially validate the descriptor extraction pipeline. Isolated leaves on a white background make segmentation trivial, so each descriptor can be checked before any variability is introduced. Its 32 species are too few to support general claims, which is why a second stage is needed.

- **Natural conditions** — Pl@ntNet-300K as the main source, with the PlantCLEF 2015–16 leaf and LeafScan subsets used for the repeatability and quality-stratified analyses, subject to the counts above.
- **Herbarium 2021** — not used now, but kept on record. It is the only dataset with a metric reference, so it remains the natural option if absolute size is studied later. The current priority is descriptors obtainable directly from leaf geometry and appearance.

Two consequences follow. Since none of the datasets in use provides a scale, the descriptors will be dimensionless or relative rather than absolute. And since none of the four provides ground-truth morphological annotations — no segmentation masks, no contour points — validation will require manually annotating a small sample.

Sources

- Flavia: <https://flavia.sourceforge.net/> — Wu *et al.* (2007), <https://arxiv.org/abs/0707.4289>
- Pl@ntNet-300K: <https://zenodo.org/records/4726653> — <https://github.com/plantnet/PlantNet-300K>
- PlantCLEF 2015: <https://www.imageclef.org/lifeclef/2015/plant> — 2016: <https://www.imageclef.org/lifeclef/2016/plant>
- PlantCLEF 2022: <https://www.imageclef.org/PlantCLEF2022> — 2024: <https://www.imageclef.org/PlantCLEF2024> — 2024 pretrained model: <https://zenodo.org/records/10848263>
- Herbarium 2021: <https://www.frontiersin.org/journals/plant-science/articles/10.3389/fpls.2021.787127/full> — <https://www.kaggle.com/c/herbarium-2021-fgvc8>